

1 Limits and Continuity

$\lim_{x \rightarrow a} f(x)$ exists $\Leftrightarrow \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$

Continuity at $x = c$ $\Leftrightarrow \lim_{x \rightarrow a} f(x) = L = f(c)$

Continuity at endpoints only one-sided limit.

Continuity at an interval $\Leftrightarrow \forall c \in I (f \text{ is continuous at } c)$

Squeeze Theorem

Let $I = (a, c) \cup (c, b)$. $\forall x \in I (f(x) \leq g(x) \leq h(x))$

$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = L \Rightarrow \lim_{x \rightarrow c} g(x) = L$

Special Limit

$$\lim_{x \rightarrow c} g(x) = 0 \Rightarrow \lim_{x \rightarrow c} \frac{\sin(g(x))}{g(x)} =$$

$$\lim_{x \rightarrow c} \frac{\tan(g(x))}{g(x)} = 1$$

Indeterminate Limit

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} \Rightarrow \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

2 Theorems

Intermediate Value Theorem

f is continuous on $[a, b]$, $k \in [f(a), f(b)] \Rightarrow \exists c \in [a, b] (f(c) = k)$

Rolle's Theorem

f is differentiable on (a, b) , $f(a) = f(b) \Rightarrow \exists c \in (a, b) (f'(c) = 0)$

Mean Value Theorem

f is differentiable on (a, b) , $\exists c \in (a, b) (f'(c) = \frac{f(b)-f(a)}{b-a})$

3 Derivative

Limit Definition

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h)-f(x_0)}{h}$$

Differentiability on an Interval $\Leftrightarrow$ Continuous on the interval

Quotient Rule

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{\frac{du}{dx} \cdot v - \frac{dv}{dx} \cdot u}{v^2}$$

Inverse Derivative

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

Derivative of Parametric

$$\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$$

$$\frac{d^2y}{dx^2} = \frac{f'(t) \cdot g''(t) - f''(t) \cdot g'(t)}{(f'(t))^3}$$

Critical Point

Not end point, $f'(c) = 0 \vee f'(c)$ does not exist.

Absolute Extrema by First Derivative

$\forall x < c (f'(x) > 0) \wedge \forall x > c (f'(x) < 0) \Rightarrow f$ has an absolute maximum at c

$\forall x < c (f'(x) < 0) \wedge \forall x > c (f'(x) > 0) \Rightarrow f$ has an absolute minimum at c

Local Extrema by First Derivative

f is differentiable on $(a, c) \cup (c, b) \wedge$ continuous at c

f' changes from + to - at $x = c \Rightarrow$ max

f' changes from - to + at $x = c \Rightarrow$ min

Second Derivative Test

$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow$ local max at c

$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow$ local min at c

4 Integration

Partial Fractions Simplify $\frac{P(x)}{Q(x)}$

Factors of $Q(x)$	
$ax + b$	$\frac{A}{ax+b}$
$(ax + b)^2$	$\frac{A}{ax+b} + \frac{B}{(ax+b)^2}$
$ax^2 + bx + c$	$\frac{Ax+B}{ax^2+bx+c}$

Trigo Substitution

Expression	Substitution
$\sqrt{a^2 - (x + b)^2}$	$x + b = a \sin \theta$
$\sqrt{(x + b)^2 + a^2}$	$x + b = a \tan \theta$
$\sqrt{(x + b)^2 - a^2}$	$x + b = a \sec \theta$

Domains: $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, -\frac{\pi}{2} < \theta < \frac{\pi}{2}, 0 < \theta < \frac{\pi}{2}$ OR $\pi \leq \theta < \frac{3\pi}{2}$

Integration by Parts

$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx$
(diff) Log, Inverse, Alge, Trig, Expo (int)

Riemann Sum

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{b-a}{n} \right) f\left(a + k\left(\frac{b-a}{n}\right)\right)$$

Fundamental Theorem of Calculus

$$\int_a^b f(x)dx = F(b) - F(a)$$

Second Fund. Theorem of Calc

$$F(x) = \int_a^x f(t)dt, a \leq x \leq b, F(a) = 0$$

$$\frac{d}{dx} \int_a^x f(t)dt = f(x)$$

Improper Integrals

$$\int_{-\infty}^{\infty} f(x)dx = \lim_{a \rightarrow -\infty} \int_a^c f(x)dx +$$

$$\lim_{b \rightarrow \infty} \int_c^b f(x)dx$$

$$\int_a^b f(x)dx = \lim_{c \rightarrow d^-} \int_a^c f(x)dx + \lim_{c \rightarrow d^+} \int_c^b f(x)dx$$

5 Integration Applications

Area between curves

$$\int_a^b |f(x) - g(x)|dx, \int_a^b |f(y) - g(y)|dy$$

Volume of Solid by Disk

$$\pi \int_a^b f(x)^2 dx - \pi \int_a^b g(x)^2 dx$$

$$\pi \int_a^b f(y)^2 dy - \pi \int_a^b g(y)^2 dy$$

Volume of Solid by Cyindrical Shell

$$2\pi \int_a^b x |f(x) - g(x)|dx$$

$$2\pi \int_a^b y |f(y) - g(y)|dy$$

Length of curve $y=f(x)$

$$\int_a^b \sqrt{1 + f'(x)^2}dx, \int_p^q \sqrt{1 + f'(y)^2}dy$$

6 Equations

Ellipse $\frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{b^2} = 1$

$x = a \cos(t) + x_0, y = b \sin(t) + y_0$

Circle $(x - x_0)^2 + (y - y_0)^2 = r^2$

$x = r \cos(t) + x_0, y = r \sin(t) + y_0$

Horizontal Hyperbola

$$\frac{(x-x_0)^2}{a^2} - \frac{(y-y_0)^2}{b^2} = 1$$

$x = a \sec(t) + x_0, y = b \tan(t) + y_0$

Vertical Hyperbola

$$\frac{(y-y_0)^2}{b^2} - \frac{(x-x_0)^2}{a^2} = 1$$

$x = a \tan(t) + x_0, y = b \sec(t) + y_0$

7 Series

Convergence of series $\lim_{n \rightarrow \infty} a_n = L$

Absolute Convergence $\sum_{n=1}^{\infty} |a_n|$ is convergent.

Geometric Series Convergent for $|r| < 1$

$$\sum_{i=1}^n ar^{i-1} = \begin{cases} \frac{a(1-r^n)}{1-r} & \text{if } r \neq 1 \\ an & \text{if } r = 1 \end{cases}$$

p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent $\Leftrightarrow p > 1$

8 Tests for Convergence

n-th Term Test

$\lim_{n \rightarrow \infty} a_n \neq 0 \vee \lim_{n \rightarrow \infty} a_n DNE \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent.

Convergence by Partial Sum

$\forall n \in \mathbb{N} \exists k \in \mathbb{R} (S_n < k) \Leftrightarrow \sum_{n=1}^{\infty} a_n$ of non-negative terms converge

Integral Test

$\forall n \in \mathbb{N} (a_n = f(n)), \forall x \geq 1 (f(x) \text{ is continuous, positive, decreasing})$

$\sum_{n=1}^{\infty} a_n$ is convergent $\Leftrightarrow \int_1^{\infty} f(x)$ is convergent

Comparison Test

$\forall n \in \mathbb{N} (0 \leq a_n \leq b_n)$

$\sum_{n=1}^{\infty} a_n$ is divergent $\Rightarrow \sum_{n=1}^{\infty} b_n$ is divergent.

$\sum_{n=1}^{\infty} b_n$ is convergent $\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent.

Ratio/Root Test

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L \text{ OR } \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L,$$

$0 \leq L < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is absolutely convergent.

$L > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent.

$L = 1$ is inconclusive.

Alternating Series Test

$$\forall n \in \mathbb{N} (b_n > 0 \wedge b_n \geq b_{n+1}) \wedge \lim_{n \rightarrow \infty} b_n = 0 \Rightarrow$$

Alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \dots \text{ and}$$

$$\sum_{n=1}^{\infty} (-1)^n b_n = -b_1 + b_2 - b_3 + b_4 - \dots \text{ are convergent.}$$

9 Power Series

Deriving Radius of Convergence:

Using Ratio or Root Test!

Radius of Convergence

$$\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = \frac{1}{R} \text{ OR } \lim_{n \rightarrow \infty} \sqrt[n]{|c_n|} = \frac{1}{R}$$

$$\text{For } \sum_{n=0}^{\infty} c_n (x-a)^n,$$

$|x-a| < R$: Absolute Convergence

$|x-a| = R$: Check with Convergence Tests

Power Series Representation

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$$

$$\int f(x) = \sum_{n=0}^{\infty} \frac{c_n (x-a)^{n+1}}{n+1} + C$$

Taylor Series of f at $x = a$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, |x-a| < R$$

Maclaurin Series of f = Taylor Series of f at $a = 0$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-a)^n, |x-a| < R$$

Common Maclaurin Series

$$\bullet \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

$$\bullet e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\bullet \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\bullet \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\bullet \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$$

10 3D Coordinate System

Distance between points

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Length of Vector

$$\|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2}$$

Dot Product

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \cos \theta$$

$$\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$$

Cross Product

Area of a parallelogram with sides $\vec{a}$ and $\vec{b}$:

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta$$

Distance from Point Q to Line PR:

$$\frac{\|\vec{PQ} \times \vec{PR}\|}{\|\vec{PR}\|}$$

Angle between Vectors

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

$$\sin \theta = \frac{\|\vec{a} \times \vec{b}\|}{\|\vec{a}\| \|\vec{b}\|}$$

Projection of $\vec{b}$ on $\vec{a}$

$$\text{Vector Projection } proj_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{\vec{a} \cdot \vec{a}} \vec{a}$$

$$\text{Scalar Projection } comp_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}$$

Distance between Point and Plane

Plane Equation: $d = ax + by + cz$, Point: (x_0, y_0, z_0)

$$Distance = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Cylinder

A surface where $\exists$ Plane P such that all the planes parallel to P intersect the surface in the same curve.

11 Calculus but 3D :D

Vector-Valued Function $f: \mathbb{R} \rightarrow \mathbb{V}_n$

Derivative of Vector-valued Function

$$r'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

Arc Length of Curve

$$\int_a^b \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2} dt = \int_a^b \|r'(t)\| dt$$

Tangent line to Curve

$r'(a)$ is the tangent vector to the curve at a .

Form a line with the origin point at $t = a$ and the tangent vector.

Differentiability of $f(x,y)$

Tangent Plane at $x = a, y = b$ is a good approximation to f at points close to (a, b)

Chain Rule but 3D

$$z = f(g(t), h(t)) \Rightarrow \frac{dz}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

$$z = f(g(s, t), h(s, t)) \Rightarrow \frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$z = f(g(s, t), h(s, t)) \Rightarrow \frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial t}$$

Increment/Differential of z

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$$

$$dz = f_x(x, y)dx + f_y(x, y)dy$$

Gradient of $f(x, y)$

$\nabla f = \langle f_x, f_y \rangle$ is normal to the level curve $f(x, y) = k$

$\nabla f = \langle f_x, f_y, f_z \rangle$ is normal to any curve C on surface

$$S \quad f(x, y, z) = k$$

Tangent Plane to Surface $z = f(x, y)$

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Tangent Plane to Level Surface

$$\nabla f(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) +$$

$$f_z(x_0, y_0, z_0)(z - z_0) = 0$$

Direction Derivative of $f(x, y)$

At (x_0, y_0) , in the direction of unit vector $\langle a, b \rangle$:

$$D_u f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

$$D_u f(x, y) = \nabla f \cdot \langle a, b \rangle$$

Maximum Rate of Change of f at point P

∇f is maximum ($\|\nabla f(P)\|$) in direction $\nabla f(P)$

∇f is minimum ($-\|\nabla f(P)\|$) in direction $-\nabla f(P)$

Critical Points of $f(x, y)$

$f_x(a, b) = 0 = f_y(a, b)$ OR f_x or f_y does not exist at (a, b)

Local Maxima, Minima, Saddle Points

$\forall (x, y)$ in a disk with center (a, b) , $f(x, y) \leq f(a, b) \Rightarrow (a, b)$ is a local maximum point, $f(a, b)$ is a local maximum value

$\forall (x, y)$ in a disk with center (a, b) , $f(x, y) \geq f(a, b) \Rightarrow (a, b)$ is a local minimum point, $f(a, b)$ is a local minimum value

Critical point, $\forall$ open disk centered (a, b) , $(x, y) \in \mathbb{D}, f(x, y) < f(a, b)$ and $(x, y) \in \mathbb{D}, f(x, y) > f(a, b) \Rightarrow (a, b)$ is a saddle point of f

Second Derivative Test for $f(x, y)$

Discriminant $D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$

$D > 0 \wedge f_{xx}(a, b) > 0 \Rightarrow$ Local Minima

$D > 0 \wedge f_{xx}(a, b) < 0 \Rightarrow$ Local Maxima

$D < 0 \Rightarrow$ Saddle Point

12 Equations but 3D :)

Line

$$\vec{r} = \vec{r}_0 + t\vec{v}, \quad t \in \mathbb{R}$$

$$x = x_0 + at, y = y_0 + bt, z = z_0 + ct, \quad t \in \mathbb{R}$$

Plane

$$\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r}_0$$

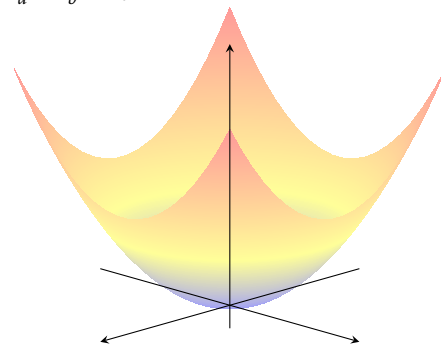
$$ax + by + cz = -(ax_0 + by_0 + cz_0) = d$$

Sphere

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2$$

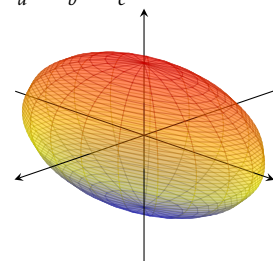
Elliptic Paraboloid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$



Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$



13 Double Pain D:

Double Riemann Sum

$$\lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

Volume of Solid above Rectangle below $z=f(x, y)$

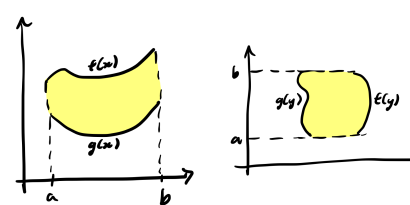
$$V = \int \int_R f(x, y) dA$$

Split Double Integral: Special Case

$$f(x, y) = g(x)h(y), \quad R = [a, b] \times [c, d]$$

$$V = \int \int_R f(x, y) dA \Rightarrow \int_a^b g(x) dx \int_c^d h(y) dy$$

Double Integral: Region Type



Type II

Type I

$$\text{Type I: } \int_a^b \int_{g(x)}^{f(x)} f(x, y) dy dx$$

$$\text{Type II: } \int_a^b \int_{g(y)}^{f(y)} f(x, y) dx dy$$

Decomposition of Domains

$$\int \int_D f(x, y) dA = \int \int_{D_1} f(x, y) dA + \dots + \int \int_{D_n} f(x, y) dA$$

Area/Volume by Iterated Integrals

$$A(D) = \int \int_D 1 dA$$

$$V(S) = \int \int \int_S 1 dV$$

Rectangle $\rightarrow$ Polar Coordinates

$$r^2 = x^2 + y^2 \Rightarrow x = r \cos \theta, y = r \sin \theta$$

If f is continuous on a polar rectangle

$$R = \{(r, \theta) : 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\} \wedge 0 \leq \beta - \alpha \leq 2\pi$$

$$\int \int_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

Surface Area of $z=f(x, y)$

$$\int \int_D dS = \int \int_D \sqrt{f_x^2 + f_y^2 + 1} dA$$

14 Ordinary Differential Equations

Reading Comprehension

Separable First Order ODE

$$\frac{dy}{dx} = f(x)g(y)$$

$$\int \frac{1}{g(y)} dy = \int f(x) dx + C$$

Non-Separable First Order ODE

Type 1:

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right)$$

Let $v = \frac{y}{x}$. Then $y' = v + xv'$.

$$v' = \frac{g(v) - v}{x} \quad \text{Type 2:}$$

$$\frac{dy}{dx} = f(ax + by)$$

Let $u = ax + by$. Then $u' = a + by'$, $y' = \frac{u' - a}{b}$.

$$\frac{u' - a}{b} = f(u) \Rightarrow u' = bf(u) + a$$

Linear First Order ODE

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\text{Let } I(x) = e^{\int P(x) dx}.$$

$$y \cdot I(x) = \int Q(x) \cdot I(x) dx$$

Bernoulli Equations

$$\frac{dy}{dx} + p(x)y = q(x)y^n, \quad n \in \mathbb{R} \setminus \{0, 1\}$$

$$\text{Let } u = y^{1-n}. \quad u' = (1-n)y^{-n}y'.$$

Multiply both sides by $(1-n)y^{-n}$.

$$(1-n)y^{-n}y' + (1-n)y^{-n}yp(x) = (1-n)y^{-n}y^nq(x)$$

$$u' + (1-n)p(x)u = (1-n)q(x)$$